

Muhammad Ahmad

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PROFESSIONAL SUMMARY

Full-Stack Engineer (MERN/Next.js) with 2+ years shipping production web applications and DeFi protocols. Experienced in scalable REST/real-time systems, MongoDB, Redis, and end-to-end delivery from API to deployment. Bonus: mainnet-deployed smart contracts, AMM architecture, and cross-chain bridge engineering on EVM chains.

EXPERIENCE

Software Engineer

NetixSol (Web3Geeks)

Apr 2025 - April 2026, Faisalabad, Pakistan

- Engineered real-time event-driven architecture using Socket.IO and Redis Pub/Sub, sustaining under 100ms message latency across 1,000+ concurrent client sessions under production load.
- Reduced gas costs by up to 80% across the full smart contract suite through storage slot packing, calldata optimization, and opcode-level refactoring in Solidity.
- Designed and deployed Next.js/Node.js/MongoDB applications with modular service architecture, supporting 1000+ concurrent users with average API response time under 100ms.
- Refactored a legacy GraphQL indexing backend (Subsquid), enforcing strict separation of concerns across resolver, service, and data-access layers — eliminating cross-layer coupling and improving long-term maintainability.

Full-Stack Developer (MERN)

CodixSol

Jul 2024 - Apr 2025, Faisalabad, Pakistan

- Improved frontend load performance by 25% across multiple client applications serving up to 1,000 daily users via code-splitting, lazy loading, and HTTP cache tuning.
- Cut average API response time by 15% by rewriting MongoDB aggregation pipelines to eliminate redundant \$lookup stages and unindexed field scans.
- Reduced total codebase size by 30% through systematic modularization and shared component abstraction, directly lowering maintenance overhead across 3+ concurrent projects.
- Mentored 2 junior developers on MERN best practices — standardized code review process, reducing recurring production defects by an estimated 40% over two release cycles.

Full-Stack Intern (MERN)

Garnica Cloud

Jan 2024 - Jul 2024, Faisalabad, Pakistan

- Built and delivered 5+ full-stack MERN applications end-to-end, contributing to a 15% reduction in internal project delivery time through reusable component systems and workflow optimization.

PROJECTS

Kaspa Finance — V3-Style Decentralized Exchange | Kasplex Blockchain (Layer 2) [app.kaspafinance.io]

- Architected a Uniswap V3-style Automated Market Maker (AMM) with tick-based concentrated liquidity — protocol reached \$15000+ peak Total Value Locked (TVL) and processed \$14700+ in single-day swap volume on mainnet.
- Engineered the full smart contract suite (factory, router, liquidity pools, staking) in Solidity with OpenZeppelin; applied gas optimization strategies, reducing per-transaction cost significantly vs. naive V2-style implementation.
- Built Next.js SSR frontend integrated via WAGMI and Viem, enabling real-time swap execution, LP position management, and staking dashboards with live on-chain state.
- Achieved comprehensive test coverage using Jest and Hardhat across core AMM invariants, tick math edge cases, and liquidity provisioning flows before mainnet deployment.

Kaspa Bridge — Cross-Chain Bridge | BSC, ETH, Base, Kasplex (Layer 2) [kaspabridge.com]

- Engineered the first-ever cross-chain bridge for Kasplex Layer 2 using Hyperlane's interoperability protocol, enabling USDT and USDC transfers across BSC, Ethereum, Base, and Kasplex L2 — bringing stablecoin liquidity to Kasplex L2 for the first time on mainnet.
- Configured and deployed Hyperlane relayer infrastructure for cross-chain message passing across 4 EVM-compatible mainnets, achieving average transaction confirmation time under 10 seconds end-to-end.
- Accumulated \$2,000+ in bridge volume within the first week of mainnet launch, validating protocol stability and cross-chain settlement correctness across BSC, Ethereum, Base, and Kasplex L2.
- Customized Hyperlane's frontend UI and smart contract integration layer, implementing real-time cross-chain transaction status tracking with live on-chain state updates via ethers.js and WAGMI.

DevPopIt — Real-Time Multiplayer Betting Game |BSC, ETH [[Live](#)]

- Architected server-authoritative real-time multiplayer game engine using Socket.IO and Node.js, supporting 1,000+ concurrent users with fully synchronized game state across all active game rooms.
- Engineered escrow-based crypto betting system accepting ETH, BNB, and stablecoins — handling wager locking, turn-based game resolution, and automated winner payout with on-chain settlement per game session.
- Implemented server-side move validation, enforcing game rules (1–3 bubble pops per turn) and loss condition detection, preventing client-side manipulation and ensuring tamper-proof game outcomes.
- Built a full-stack MERN application with React frontend delivering sub-second UI state sync on every player action via event-driven Socket.IO architecture, serving 1,000+ registered users on mainnet.

EDUCATION & CERTIFICATES

Full-Stack MERN Developer — Certification

Gamica Cloud | 2024 — Completed structured full-stack curriculum; received direct internship offer upon completion

Intermediate — Economics

B.I.S.E Faisalabad | 2019 – 2021

SKILLS

Languages: JavaScript, TypeScript, Python, Solidity

Frontend: React, Next.js (Server-Side Rendering / Static Site Generation), Tailwind CSS, Web3.js

Backend: Node.js, Express.js, NestJS, GraphQL, REST API, Microservices

Databases: MongoDB, PostgreSQL, Redis

Blockchain: Solidity, EVM, ethers.js, WAGMI, Viem, The Graph, OpenZeppelin, Subsquid, Uniswap V2/V3 SDK, Layer 2 (L2), Smart Contract

Security Tools & Infra: Docker, CI/CD, Git, Hardhat, Foundry, Vercel, Agile